

Viswa Subramanian

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SUMMARY: Robotics & AI Engineer specializing in Computer Vision, Deep Learning, and Perception Systems. Experienced in developing multi-task learning architectures for real-time vision-based tasks such as object detection, segmentation, pose estimation. Strong background in SLAM, sensor fusion, and embodied AI deployment on edge platforms.

EDUCATION

UNIVERSITY OF CALIFORNIA SAN DIEGO, CA, USA

Sep 2024 - Jun 2026*

Master of Science, Mechanical and Aerospace Engineering (Robotics & AI)

Courses: Deep Learning, Deep Generative AI, Sensing and Estimation in Robotics, Advanced Computer Vision

PSG COLLEGE OF TECHNOLOGY, TAMIL NADU, INDIA

Oct 2020 - May 2024

Bachelor of Engineering, Robotics and Automation Engineering

Courses: Calculus, Linear Algebra, Probability & Statistics, Data Structures, Machine Learning, AI for Robotics

SKILLS

AI Focus : Hyperparameter Tuning, Model Optimization, Edge AI Deployment, Multimodal AI Models, Diffusion Models

Robotics Focus : Pose Estimation, Sensor Fusion, Depth Estimation, 3D Reconstruction, SLAM, Extended Kalman Filter

Languages & Tools: Python, C, C++, MATLAB, LATEX, Git, Linux, Docker, ROS, Gazebo

Libraries & Frameworks: Numpy, Pandas, Matplotlib, OpenCV, PyTorch, TensorFlow, JAX, ONNX, TensorRT

Hardware : LiDAR, Intel Realsense, OakD; UR5, Kinnova, FANUC arms; Spartron, STIM IMUs; Arduino, Raspberry Pi

Soft Skills : Effective Communication, Teamwork, Critical Thinking, Problem Solving, Attention to detail, Adaptability

EXPERIENCE

Coordinated Robotics Lab, UCSD, San Diego - (Sponsored by GTI Energy)

Jun 2025 - Mar 2026

Graduate Researcher | Prof. THOMAS R BEWLEY, UCSD

- Engineered a multi-task MobileNetV2 CNN from scratch achieving **98%** classification accuracy and $\leq 2.8^\circ$ MAE angle error
- Designed a **circular regression head** for continuous joint angle estimation, eliminating the wrap-around discontinuity
- Exported trained model to **ONNX FP32 (11.5 MB)** and deployed on Raspberry Pi 5 achieving **~45ms** per inference

CRAS, INESC TEC, University of Porto, Portugal

Jan 2024 - May 2024

Research Intern | Prof. NUNO A CRUZ, Coordinator of CRAS

- Developed **Linux-based C++ drivers** for a new tactical-grade IMU integrated into an autonomous surface vehicle
- Conducted sensor calibration and drift analysis achieving gyroscope **bias stability of 2°** to ensure high-fidelity localization

PROJECTS

Uncertainty-Aware Diffusion based SDF Completion for Robot Safety

- Replaced DDPM with Flow Matching in Diffusion-SDF, achieving **19.4x faster sampling** in Diffusion-SDF faster 3D shape completion, enabling uncertainty estimation from partial point cloud observations trained on **ShapeNet/Acronym**
- Reproduced core shape completion quality metrics at significantly reduced compute **Nvidia RTX 2080 Ti** instead of **A100**

Experimenting Visual-Inertial SLAM Vs LiDAR SLAM in a Hexapod for Uneven Terrain

- Implementing visual-inertial and LiDAR simultaneous localization & mapping (SLAM)
- Currently integrating stereo cameras and Lidar, performing sensor calibration and real-time testing on physical platforms

Unified framework for Video Analysis

- Created responsive **PyQt5 GUI** to upload videos & perform depth estimation, segmentation, object detection and tracking
- Optimized inference speed by **40% via FP16** quantization, enabling stable **30+ FPS** across MiDaS, Mask R-CNN, YOLO

Underwater Waste Classification using Blue ROV (Bachelor Thesis)

- Curated datasets with data augmentation (rotations, flips, lighting adjustments) to increase dataset diversity, model robustness
- Deployed the final model on a BlueROV for real-time underwater waste identification with **0.70 mAP** classification accuracy

Other Projects

- Deep learning experimentation - Hyperparameter tuning for models like FCN, RNN, ViT for better performance
- ConvNeXt Norm - Rebuilt ConvNeXt from scratch, comparing LayerNorm vs Dynamic Tanh normalization
- SLAM Problem - Execute robot localization and occupancy grid mapping using Lidar measurements with some latency
- Generative AI - Implemented VAE, GANs for high-fidelity medical X-ray data generation using PyTorch

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projects. I look forward to applying my strengths in CV model design, training, and deployment to support your product and research goals.